A proof of the conjectured recurrence for OEIS A213203

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Abstract

OEIS A213203 carries a conjectured linear recurrence with polynomial coefficients, of order 4. The entry posts no generating function, so the usual route through a functional equation is unavailable. It does post a closed form for $a(n)$, and that is enough: the closed form is a sum of finitely many hypergeometric terms, and for such a term every shift quotient is a rational function of n. The conjectured recurrence therefore reduces to two rational-function identities, one for each similarity class of terms, and each is decided exactly by cancellation. No expansion is truncated and nothing is guessed; the procedure returns a proof or a refutation. Here it returns a proof, valid for every n outside an explicitly computed finite set.

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1 The sequence and the conjecture

OEIS A213203 is "The sum of the first n! integers, with every n-th integer taken as negative". It has offset 1 and begins

$$a(1), \ldots, a(8) \;=\; -1, -1, 3, 132, 4260, 172440, 9069480, 609618240, \;\ldots$$

The entry states, not as a conjecture,

a(n) = n * (n-1)! * ((n-2)*(n-1)! - 1)/2.

that is,

$$a(n) \;=\; \frac{n\,((n-2)\,(n-1)! - 1)\,(n-1)!}{2}. \tag{1}$$

This is the only input the proof takes from the entry, and it was checked against the entry's own DATA before use.

The entry separately carries the following comment:

Conjecture: a(n) + (-n^2-n-11)*a(n-1) + (n^3+7*n^2-13*n+39)*a(n-2) - 2*(n-2)*(4*n^2-2*n-15)*a(n-3) + 20*(n-2)*(n-3)*(n-4)*a(n-4) = 0. - *R. J. Mathar*, Mar 21 2013

As of the "Last modified" line on the live entry (2013-03-21, revision 23) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Written out, the conjecture is

$$1\,a(n)+(-n^2-n-11)\,a(n-1)+(n^3+7n^2-13n+39)\,a(n-2)-2\,(n-2)\left(4n^2-2n-15\right)\,a(n-3)+20\,(n-4)\,(n-3 \tag{2}$$

2 Hypergeometric terms

Definition 1. A nonzero expression $c(n)$ is a *hypergeometric term* in n if the quotient $c(n+1)/c(n)$ is a rational function of n. Two hypergeometric terms c and d are *similar*, written $c \sim d$, if $c(n)/d(n)$ is a rational function of n.

Similarity is an equivalence relation on hypergeometric terms. Products of factorials, binomial coefficients with arguments linear in n, exponentials z^n and polynomials are hypergeometric, and so is any product of these; a sum of them need not be.

Lemma 2. *If c is a hypergeometric term then for every integer $i \geq 0$ the shift quotient $c(n-i)/c(n)$ is a rational function of n, computable from $c(n+1)/c(n)$ by a finite number of substitutions and multiplications.*

Proof. Write $\rho(n) = c(n+1)/c(n)$, a rational function by hypothesis. Then $c(n-1)/c(n) = 1/\rho(n-1)$, and inductively $c(n-i)/c(n) = \prod_{j=1}^{i} \rho(n-j)^{-1}$, a finite product of rational functions. □

The fact that makes the test complete rather than merely sufficient is the linear independence of dissimilar terms. It is classical; see Petkovšek, Wilf and Zeilberger [1], Chapter 5.

Lemma 3. *Let $c_1, \ldots, c_m$ be hypergeometric terms, pairwise dissimilar. If $\sum_{j=1}^{m} u_j(n)c_j(n) = 0$ for rational functions u_j and all large n, then every u_j is the zero rational function.*

3 The recurrence as a finite set of rational identities

Proposition 4. *Let $a(n) = \sum_{j=1}^{m} c_j(n)$ with each c_j hypergeometric, and let $p_0, \ldots, p_r$ be polynomials. Group the c_j into similarity classes and write each class as $g_\ell(n)\, f_{\ell,s}(n)$ for a representative g_ℓ and rational functions $f_{\ell,s}$. Put*

$$S_\ell(n) \;=\; \sum_{i=0}^{r} \sum_{s} \, p_i(n)\, f_{\ell,s}(n-i)\, \frac{g_\ell(n-i)}{g_\ell(n)},$$

a rational function of n by Lemma 2. Then

$$\sum_{i=0}^{r} p_i(n)\, a(n-i) \;=\; \sum_{\ell} g_\ell(n)\, S_\ell(n),$$

and the recurrence $\sum_i p_i(n)a(n-i) = 0$ holds for all large n if and only if $S_\ell = 0$ for every ℓ.

Proof. The displayed identity is the definition of S_ℓ rearranged: each $a(n-i)$ is the sum of its terms, each term is its class representative times a rational function, and the representatives are pulled out. The representatives g_ℓ are pairwise dissimilar by construction, so Lemma 3 applies to the right-hand side: it vanishes for all large n exactly when every S_ℓ vanishes identically. Sufficiency is immediate. □

Proposition 4 turns the conjecture into a finite computation in $\mathbb{Q}(n)$, with no truncation and no search. The only n it says nothing about are those where some S_ℓ has a pole or some $c_j(n-i)$ is undefined, and these are finite in number and listed below.

4 The computation for A213203

The closed form (1) splits into two similarity classes, with representativees

$$\frac{n^2\,(n-1)!^2}{2}, \qquad -\frac{n\,(n-1)!}{2}$$

Carrying out Proposition 4 with the coefficients of (2) gives

$$S_1(n) \;=\; 0, \qquad S_2(n) \;=\; 0$$

so every class residual vanishes identically and the conjecture follows.

Theorem 5. *For all $n > 4$,*

$$1\,a(n)+(-n^2-n-11)\,a(n-1)+(n^3+7n^2-13n+39)\,a(n-2)-2\,(n-2)\left(4n^2-2n-15\right)\,a(n-3)+20\,(n-4)\,(n-3$$

Proof. Immediate from Proposition 4 and the computation above, the excluded values being those at which a class residual has a pole or a term of (1) is undefined. □

5 Verification

Two checks were made, and both are independent of the algebra above.

First, the closed form (1) was evaluated at the first 13 indices and compared term by term with the entry's DATA; the values agree exactly. A formula that did not reproduce the entry's own terms would not have been used.

Second, the conjectured recurrence (2) was evaluated on the published terms in exact integer arithmetic at every index where it applies: 11 instances beginning at $n = 5$, all of them zero. This is not part of the proof, which is symbolic, but it would have caught a transcription error in the coefficients.

References

[1] M. Petkovšek, H. S. Wilf and D. Zeilberger, *A=B*, A K Peters, Wellesley MA, 1996.

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A213203, 2013-03-21.

[3] R. L. Graham, D. E. Knuth and O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison-Wesley, 1994.